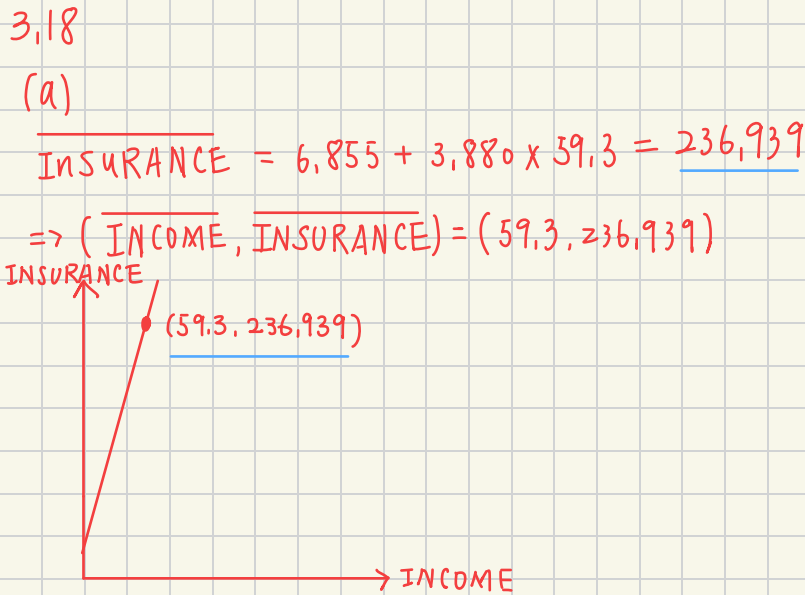


- 3.18** A life insurance company examines the relationship between the amount of life insurance held by a household and household income. Let *INCOME* be household income (thousands of dollars) and *INSURANCE* the amount of life insurance held (thousands of dollars). Using a random sample of $N = 20$ households, the least squares estimated relationship is

$$\widehat{INSURANCE} = 6.855 + 3.880 INCOME$$

(se) (7.383) (0.112)

- Draw a sketch of the fitted relationship identifying the estimated slope and intercept. The sample mean of *INCOME* = 59.3. What is the sample mean of the amount of insurance held? Locate the point of the means in your sketch.
- How much do we estimate that the average amount of insurance held changes with each additional \$1000 of household income? Provide both a point estimate and a 95% interval estimate. Explain the interval estimate to a group of stockholders in the insurance company.



(b)

$b_2 = 3.880$, degree of freedom is $20 - 2 = 18$

→ the 95% CI is $[b_2 - t_{0.975, 18} \times se(b_2), b_2 + t_{0.975, 18} \times se(b_2)]$

$t_{0.975, 18} = 2.101$

$$\Rightarrow [3.880 - 2.101 \times 0.112, 3.880 + 2.101 \times 0.112]$$

$$\Rightarrow \underline{[3.6447, 4.1153]} \text{ 單位: 千元}$$

家庭收入每±增加1000\$, 平均壽險約±增加3880\$, 在重覆約有95%的結果平均壽險增加在3644.7 ~ 4115.3之間。

- c. Construct a 99% interval estimate of the expected amount of insurance held by a household with \$100,000 income. The estimated covariance between the intercept and slope coefficient is -0.746 .
- d. One member of the management board claims that for every \$1000 increase in income the average amount of life insurance held will increase by \$5000. Let the algebraic model be $INSURANCE = \beta_1 + \beta_2 INCOME + e$. Test the hypothesis that the statement is true against the alternative that it is not true. State the conjecture in terms of a null and alternative hypothesis about the model parameters. Use the 5% level of significance. Do the data support the claim or not? Clearly, indicate the test statistic used and the rejection region.
- e. Test the hypothesis that as income increases the amount of life insurance held increases by the same amount. That is, test the null hypothesis that the slope is one. Use as the alternative that the slope is larger than one. State the null and alternative hypotheses in terms of the model parameters. Carry out the test at the 1% level of significance. Clearly indicate the test statistic used, and the rejection region. What is your conclusion?

$$(c) \alpha = 0,01, \text{ cov}(b_1, b_2) = -0,746, b_1 = 6,855, b_2 = 3,880$$

$$\widehat{INSURANCE} = 6,855 + 3,880 \times 100 = 394.855 \text{ (千元)}$$

$$\begin{aligned} SE(b_1 + 100b_2) &= \sqrt{\text{Var}(b_1) + 100^2 \text{Var}(b_2) + 2 \times 100 \times \text{cov}(b_1, b_2)} \\ &= \sqrt{7,383^2 + 100^2 \times 0,112^2 + 2 \times 100 \times -0,746} \\ &= \sqrt{54,308689 + 125,44 - 149,2} \\ &= \sqrt{30,748689} \approx 5,545 \end{aligned}$$

$$99\% \text{ CI is } [\widehat{INSURANCE} - t_{0,995,18} \times SE(\widehat{INSURANCE}), \widehat{INSURANCE} + t_{0,995,18} \times SE(\widehat{INSURANCE})]$$

$$\begin{aligned} \frac{t_{0,995,18}}{= 2,878} &= [394,855 - 2,878(5,545), 394,855 + 2,878(5,545)] \\ &= [378,897, 410,813] \text{ (千元)} \end{aligned}$$

$$(d) H_0: \beta_2 = 5, H_1: \beta_2 \neq 5$$

$$t = \frac{b_2 - 5}{SE(b_2)} = \frac{3,880 - 5}{0,112} = -10$$

$$\alpha = 0,05 \Rightarrow \text{rejection region is } |t| > t_{0,975,18} = 2,101$$

$$\because |t| = 10, \text{ Hence the result reject } H_0$$

$$(e) H_0: \beta_2 = 1 \quad H_1: \beta_2 > 1$$

$$t^* = \frac{b_2 - 1}{se(b_2)} = \frac{3,880 - 1}{0,112} = 25,7143$$

$\alpha = 0,01$, $df = 18 \Rightarrow$ the rejection region is $t > t_{0,99,18} = 2,552$

$\therefore t^* > 2,552 \quad \therefore$ the result reject H_0

3.23 The data file *collegetown* contains data on 500 single-family houses sold in Baton Rouge, Louisiana, during 2009–2013. The data include sale price in \$1000 units, *PRICE*, and total interior area in hundreds of square feet, *SQFT*.

- Using the quadratic regression model, $PRICE = \alpha_1 + \alpha_2 SQFT^2 + e$, test the hypothesis that the marginal effect on expected house price of increasing the size of a 2000 square foot house by 100 square feet is less than or equal to \$13000 against the alternative that the marginal effect will be greater than \$13000. Use the 5% level of significance. Clearly state the test statistic used, the rejection region, and the test p -value. What do you conclude?
- Using the quadratic regression model in part (a), test the hypothesis that the marginal effect on expected house price of increasing the size of a 4000 square foot house by 100 square feet is less than or equal to \$13000 against the alternative that the marginal effect will be greater than \$13000.

Use the 5% level of significance. Clearly state the test statistic used, the rejection region, and the test p -value. What do you conclude?

- Using the quadratic regression model in part (a), estimate the expected price $E(PRICE|SQFT) = \alpha_1 + \alpha_2 SQFT^2$ for a house of 2000 square feet. Construct a 95% interval estimate of the expected price. Describe your interval estimate to a general audience.
- Locate houses in the sample with 2000 square feet of living area. Calculate the sample mean (average) of their selling prices. Is the sample average of the selling price for houses with $SQFT = 20$ compatible with the result in part (c)? Explain.

3.23

```
## 3.23(a)
## test statistic is: t = (alpha2_hat - 0.325)/se(alpha2_hat)
## rejection region: tc > 1.647919
## test p-value : -26.72875
## hence t* is falls in the non-rejection region
```

```
## 3.23(b)
## test statistic is: t = (alpha2_hat - 0.1625)/se(alpha2_hat)
## rejection region: tc > 1.647919
## test p-value : 4.189467
## hence t* is falls in the non-rejection region
```

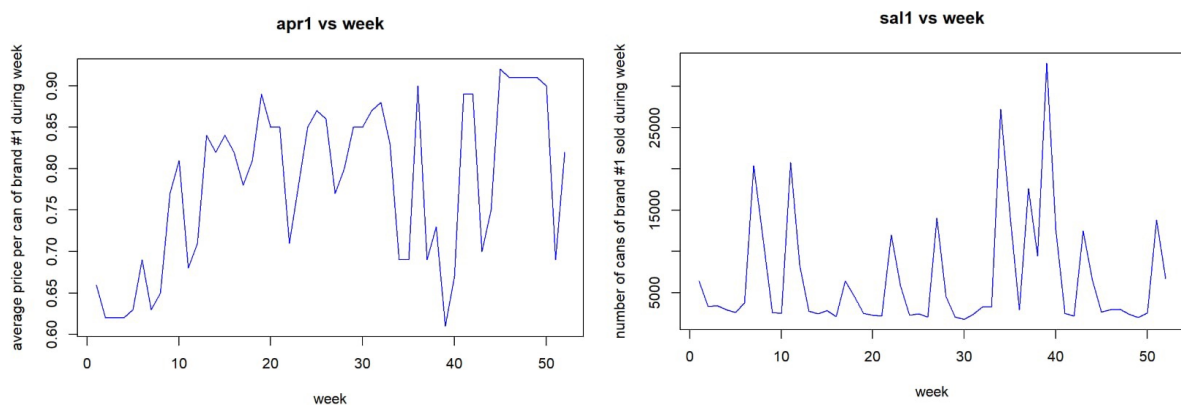
```
## 3.23(c)
## expected price is 167.3735
## 95% CI of the expected price: [158.0481,176.6988]
## In repeated sampling,95% intervals constructed this way will contain true value
## of expected price
```

```
## 3.23(d)
## In the sample there are 3 houses with 2000 square feet
## and sold 138,169,183
## The sample mean is 163.3333
## and it is inside the interval we estimate
```

3.31 Data on weekly sales of a major brand of canned tuna by a supermarket chain in a large midwestern U.S. city during a mid-1990s calendar year are contained in the data file *tuna*. There are 52 observations for each of the variables. The variable *SAL1* = unit sales of brand no. 1 canned tuna, and *APR1* = price per can of brand no. 1 tuna (in dollars).

- Calculate the summary statistics for *SAL1* and *APR1*. What are the sample means, minimum and maximum values, and their standard deviations. Plot each of these variables versus *WEEK*. How much variation in sales and price is there from week to week?
- Plot the variable *SAL1* (y-axis) against *APR1* (x-axis). Is there a positive or inverse relationship? Is that what you expected, or not? Why?
- Create the variable $PRICE1 = 100APR1$. Estimate the linear regression $SAL1 = \beta_1 + \beta_2 PRICE1 + e$. What is the point estimate for the effect of a one cent increase in the price of brand no. 1 on the sales of brand no. 1? What is a 95% interval estimate for the effect of a one cent increase in the price of brand no. 1 on the sales of brand no. 1?
- Construct a 90% interval estimate for the expected number of cans sold in a week when the price per can is 70 cents.
- Construct a 95% interval estimate of the elasticity of sales of brand no. 1 with respect to the price of brand no. 1 "at the means." Treat the sample means as constants and not random variables. Do you find the sales are fairly elastic, or fairly inelastic, with respect to price? Does this make economic sense? Why?
- Test the hypothesis that elasticity of sales of brand no. 1 with respect to the price of brand no. 1 from part (e) is minus three against the alternative that the elasticity is not minus three. Use the 10% level of significance. Clearly, state the null and alternative hypotheses in terms of the model parameters, give the rejection region, and the *p*-value for the test. What is your conclusion?

3.31(a)



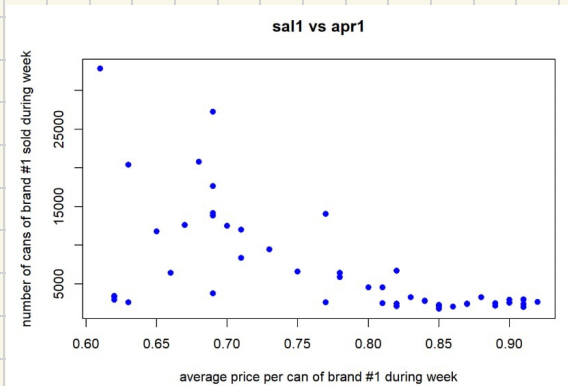
##	variable	N	mean	Std. dev.	min	max
## 1	sal1	52	6718.7115	6.915083e+03	1762.00	32820.00
## 2	apr1	52	0.7825	9.803711e-02	0.61	0.92

3.31(a)

There is an inverse relationship between the weekly sales and weekly price.

weekly sales exhibit much greater variation than weekly price.

(b)



3.31(b)

There is an inverse relationship between the weekly sales and weekly price.

it is As expected, the higher the price, the lower the sales volume.

(c)

3.31(c)

a one-cent increase in the price of brand one will reduce the expected

weekly sales of brand one tuna by 434.4473 cans

and 95% CI estimate is [-592.2897, -276.6049]

(d)

3.31(d)

The 90% CI estimate is [8624.934, 11980.87]

(e)

3.31(e)

estimate of the elasticity is -5.059825

The 95% CI estimate is [-6.898149, -3.221501]

since absolute value of elasticity is 5.059825

this is elastic and hence the result satisfied Law of Demand and Supply

(f)

3.31(f)

the rejection region is $t_c > 1.675905$ or $t_c < -1.675905$

and test p-value is -2.250572

hence We reject the null hypothesis

$$\therefore E = \beta_2 \times \frac{\overline{APR1}}{\overline{SAL1}} \Rightarrow H_0: E = -3, H_1: E \neq -3$$